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By Stanford ML Group

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Tech Overview

BRL Architecture

T-RO, IJRR

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Summary

The paper proposes a knowledge-based POMDP framework that decides when to ask for help and what to ask by maintaining a belief over symbolic world hypotheses, triggering queries when belief entropy is high, and selecting state-level yes/no queries that maximize expected entropy reduction. A POMCP-based planner is integrated with a centralized symbolic knowledge base and a feedback manager that translates predicates into user-facing questions. Experiments in simulated waste sorting and a tomato-harvesting setup (with an additional WoZ user study) show that the approach can retain high task success while substantially reducing the frequency of user queries compared to continuous supervision.



Strengths

Technical novelty and innovation

- The paper presents a clear integration of symbolic task models (STRIPS-style preconditions/effects) with online POMDP planning via a particle-like “frontier over reachable successor states,” a pragmatic compromise between fully factored beliefs and global particle filtering.
- The what-to-ask component is principled (expected entropy reduction over hypotheses) and tightly coupled to symbolic action feasibility, resulting in informative state-level queries rather than action-level ones.
- The architecture cleanly unifies sensing, planning, user feedback, and a shared knowledge base, with predicates used both for planning and interaction.

Experimental rigor and validation

- Ablations over confidence thresholds (τ) systematically show trade-offs between success and query load across two domains of differing complexity.
- Baseline comparisons include Always-Ask, No-Ask, Random, and an action-query baseline (KnowNo), highlighting the distinct advantages and failure modes of state-level vs. action-level queries.
- A scalability test (4→6 objects) surfaces computational effects and selective querying properties as the symbolic state space grows.

Clarity of presentation

- The system architecture, planning loop, belief update, and query-selection pipeline are well-described with intuitive figures and algorithmic summaries.
- Related work is broad and structured, with a useful comparison table positioning the contribution along axes of interpretability, action models, uncertainty-aware planning, and feedback.

Significance of contributions

- The work targets an important HRI question—deciding when and what to query—to reduce operator workload without sacrificing task performance.
- Evidence that state-level queries can help maintain high success while substantially lowering query rates relative to dense supervision is valuable for practical deployments.

Weaknesses

Technical limitations or concerns

- The belief approximation is highly myopic: particles are limited to frontier states induced by the current committed knowledge state and the next action, effectively collapsing the global belief after each step. This can induce overconfidence, lose multi-step dependencies, and depart from standard POMDP inference.
- Observation models and their calibration are domain-specific and delegated to an appendix, yet they critically determine likelihoods and thus both the confidence and query decisions; limited analysis is provided on their accuracy or robustness to sensor/model bias.
- The entropy-based confidence trigger and one-step query objective do not explicitly optimize downstream task value (e.g., expected value of information), so there is no guarantee that reduced entropy translates to better returns under long-horizon dependencies, especially in the tomato domain.

Experimental gaps or methodological issues

- The KnowNo baseline is action-level and may be disadvantaged by not receiving sufficient state exposure; its configuration is only briefly outlined. The large discrepancy between system-level and WoZ user results (KnowNo poor in system results but best in tomato WoZ success) suggests potential mismatch in how baselines were instantiated/contextualized in the two evaluations.
- The system-level experiments use an oracle user; the approach is not stress-tested against human labeling errors or latency, which are common in practice.
- The user study is a small-sample, WoZ-style evaluation with one exposure per condition and domain; results are mixed for SAGAT and workload is generally low across conditions, limiting strong conclusions about human factors.
- Real-robot validation is only briefly mentioned; it is unclear whether the reported success/queries were measured on hardware under closed-loop execution with the same pipeline or merely feasibility was demonstrated.

Clarity or presentation issues

- Some core implementation choices (e.g., exact observation-likelihood forms, how duplicate successor states are merged, and effects of wrong human answers) are deferred to appendices; a concise summary in the main text would improve reproducibility.
- Minor typos in formulas (e.g., reusing the same symbol in the normalization step) and notation inconsistencies appear but do not impede understanding.

Missing related work or comparisons

- The decision-theoretic VOI and p -POMDP literature (belief-dependent rewards for information gathering) is relevant for principled “what to ask” under long horizons; only entropy reduction is used here, with no comparison to VOI-based policies.
- Prior interactive/active perception and human-in-the-loop POMDP works that reason explicitly about information value and query costs could be discussed as alternative query-selection criteria and planning formulations.

Detailed Comments

Technical soundness evaluation

- The frontier-based belief approximation is practical but departs substantially from standard posterior maintenance. By committing to the MAP frontier state once confidence exceeds τ , the method implicitly prunes alternative global hypotheses that might be revived by later observations. This is acceptable as a heuristic but should be discussed as such, with analysis of failure cases (e.g., early mis-commitment cascading into later dead-ends).
- The query objective is standard expected entropy reduction over a binary hypothesis; it is sensible and cheap to compute but remains a myopic proxy. Integrating VOI or belief-dependent reward optimization (e.g., p -POMDP) would better align “what to ask” with task value under uncertainty.

- Restricting POMCP to symbolically applicable actions is good practice and likely responsible for efficiency gains; however, the runtime escalation in the 6-object tomato setting ($\sim 16\times$ per-step time) suggests more aggressive search control (e.g., progressive widening, DESPOT-style scenario sampling, caching/reuse across steps) may be needed for scale.

Experimental evaluation assessment

- The τ -ablation is thorough and domain-sensitive (threshold effects differ markedly between waste and tomato). The conclusion that “more queries is not always better” is convincingly supported.
- Baseline outcomes make sense qualitatively: Always-Ask is strong but costly; No-Ask fails; Random helps in short-horizon waste but not in long-horizon tomato; Ours strikes a useful balance. Yet, the KnowNo configuration likely needs more careful equalization of information provided to the user/LLM to ensure fairness, given its sensitivity to context.
- The scalability experiment reveals important computational bottlenecks (episode and step times, belief-frontier growth), and it is commendable that these costs are reported. Additional profiling (tree size vs. frontier size vs. observation likelihood cost) would clarify where to optimize.
- The WoZ user study is appropriately presented as preliminary. Given the mixed SAGAT and small N, the claims about user burden should be phrased cautiously (the paper is generally careful here). Evaluating resilience to occasional incorrect human answers and to response delays would strengthen external validity.

Comparison with related work (using the summaries provided)

- Online POMDP solvers such as DESPOT and POMCPOW, as well as variants handling belief-dependent rewards (p -POMCP/ p POMCPOW), provide theoretically grounded ways to optimize information acquisition. While this paper pursues an architectural/system integration angle, acknowledging and contrasting to belief-reward approaches (VOI, entropy gain as reward) would situate the “what to ask” decision more rigorously.
- Recent finite-time analyses for POMCPOW-style planners and approaches that reuse historical planning information could offer avenues to tame the observed runtime growth.

- The paper's stance relative to active visual search and interactive POMDPs could be sharpened: here, queries are human-facing predicates; in AVS, queries are (robotic) viewpoint/control choices. Both are information gathering under uncertainty; connecting to that literature could broaden the impact.

Discussion of broader impact and significance

- The paper tackles a consequential problem for practical autonomy—reducing supervision burden without sacrificing performance—via interpretable state-level queries. This interaction modality can indeed make the human's task easier than selecting among opaque action candidates.
- Risks include over-reliance on ad-hoc observation models and potential overconfidence from stepwise belief commitments; if deployed, systems should monitor for failure modes and possibly incorporate safeguards (e.g., periodic revalidation, backtracking mechanisms).
- The architecture's interpretability (exposing predicates as the locus of queries) is a strong point for transparency and post-hoc debugging.

Questions for Authors

1. How sensitive are the results to misspecified observation likelihoods $O(o|s,a)$? Could you report calibration curves or an ablation showing performance under biased or noisier observation models?
2. What happens when human feedback is wrong or delayed? Do you re-ask, degrade confidence, or allow backtracking if later evidence contradicts a committed knowledge state?
3. How large can the frontier become in practice, and how do you cap or prune it? Can you provide statistics on unique successor merging and its effect on runtime and success?
4. Could you compare your entropy-based query selection to a VOI-style criterion that explicitly considers downstream task value (e.g., via a short lookahead or a belief-dependent reward surrogate)?

5. How was the KnowNo baseline instantiated in system experiments vs. the WoZ study, and what context was provided to it? Can you reconcile the large difference in its performance across the two evaluations?
6. In the real-robot implementation, what quantitative results (success, queries, runtime) can you report under closed-loop operation? If the main numbers are from controlled environments, what gaps remain for on-hardware deployment?
7. Have you tried progressive widening or DESPOT-style scenario sampling to curb the growth in episode/step time for larger scenes? If so, how did they affect performance and query behavior?

Overall Assessment

This paper offers a well-engineered integration of symbolic knowledge, online POMDP planning, and state-level human queries that are both interpretable and informative. The architectural clarity, principled entropy-based query selection, and cross-domain evaluation are commendable, and the preliminary user study suggests the interface is usable without excessive burden. On the other hand, the belief approximation is myopic and may induce overconfidence; observation models are central yet under-analyzed; the user study is limited in scope; and fairness/details of the action-level baseline merit closer scrutiny. The work's novelty is moderate—expected-entropy queries and POMCP are established—but the system-level synthesis and explicit comparison of state- vs. action-level queries provide useful insights. With additional analysis on observation-model robustness, more rigorous/fair baseline configuration, and some algorithmic measures to address scalability and long-horizon VOI, the contribution would be strengthened. As it stands, I view this as a solid systems paper with practical value and a clear interaction design, though not yet at the level of methodological depth typically required for T-RO/IJRR. I recommend borderline acceptance if positioned as a systems/architecture contribution with expanded empirical rigor; otherwise, a revise-and-resubmit focusing on robustness, fairness, and scalability would be appropriate.

Note: Reviews are AI generated and may contain errors. Please use them as guidance and apply your own judgment.

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